CSA5111 - CNS

APPLICATION - BASED QUESTION

Assessment TOOL - 1 - CO1

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1. Scenario event communication Security. A student Organization is preparing to share Confidential Events details with its core members using a classical Encryption Method.

Caeser cipher:

The given plaintext is MEET AT NOON and the Encryption is shift is 4 position

a) Reconstruct the Encrypted message using the shift of 4 position

The Encryption formula is

$$E(x) = P \times K \bmod 26$$

plain text : MEET AT NOON

Ciphertext : Q11X EX RSSR

Final Encrypted : Q11X EX RSSR

b) Explain how an authorised member retrives the original message

↳ An authorized member who knows the Encryption shift can decrypt the message

by shifting each ciphertext letter 4 position backward in the alphabet

$$D(x) = (x-k) \bmod 26$$

D(x) - decrypted letter value

x - cipher text

k = 4 = shift value

A B C D E F G H I J K L M

N O P Q R S T U V W X Y Z

Shift 4 value backward

Q → M
I → E
I → E
X → T
E → A
X → T
R → N
S → O
S → O
R → N

∴ The Decrypted message

MEET AT NOON

2. Scenario Intercepted Suspicious Message : A Cybersecurity intern intercepts the coded Message "GSRH RH Z HVXIVG" during network Monitoring

Atbash cipher

a) Deduce the most likely original message

Given ciphertext : GSRH RH Z HVXIVG

Atbash cipher is a classical substitution cipher in which the alphabet is reversed

A is replaced by Z . . . .

Z	Y	X	V	U	T	S	R	Q	P	O
A	B	C	D	E	F	G	H	I	J	K
N	M	L	K	J	I	H	G	F	E	D
L	M	N	O	P	Q	R	S	T	U	V

C	B	A
X	Y	Z

Ciphertext → Decrypted Text

GSRH → THIS

RH → IS

Z → A

HVXIVG → SECRET

∴ Original message is THIS IS A SECRET

b) Justify the answer using repeating pattern and cipher characteristics

The Decoded message is THIS IS A SECRET

→ Atbash uses a fixed substitution produces a grammatically meaningful English sentence.

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3. Scenario: Secure Messaging System: A company deploys key-word based encryption technique for internal communication. An Employee sends the word "HELLO" using the keyword "KEY" to ensure Confidentiality

Vigenère cipher

plaintext : HELLO

key : KEY

a) Demonstrate the Encryption process step by step

$$C_i = (P_i + K_i) \bmod 26$$

plaintext : HELLO

keyword : KEYKE

plaintext	(Pi) Position	keyword	(Ki) Position
H	7	K	10
E	4	E	4
L	11	Y	24
L	11	K	10
O	14	E	4

$\Rightarrow (7 + 10) \bmod 26 = 17 \rightarrow R$

$\Rightarrow (4 + 4) \bmod 26 = 8 \rightarrow I$

$\Rightarrow (11 + 24) \bmod 26 = 9 \rightarrow J$

$\Rightarrow (11 + 10) \bmod 26 = 21 \rightarrow V$

$\Rightarrow (14+4) \bmod 26 = 18 \rightarrow S$

∴ Encrypted message is RUVS

b) Explaining how the repeating keyword Enhancing Each Characters of CT.

→ Each keyword letter produces a different shift. Therefore, the same plaintext can produce different ciphertext letters when it is combined with different keyword letters

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Scenario Suspicious Digital Asset Investigation: During a forensic audit, Investigators Suspects that a company logo file may contain hidden confidential information embedded with in it.

) Detecting hidden Data in an image.

↳ Image steganography is the technique of hiding secret information inside an

ordinary image

Investigators can detects hidden data using following methods.

(1) metadata analysis - check images and its properties, metadata for unusual changes

(2) File analysis - Examine file size and structure for suspicious data

(3) LSB analysis - Analyze least significant bits of pixels for hidden information

(4) Statistical analysis - Identify unusual patterns

(b) Two techniques to improve security:

(1) ENCRYPTION - Encrypt the message using AES before hiding it in the image

(2) SECURE STEGNOGRAPHY - use a suitable steganographic technique that minimize detectable image changes and makes hidden data to discover harder.

5. Scenario Legacy encryption in Corporate Communication : A legacy system in a company still uses a transposition - based encryption method.

a) Decryption of the ciphertext

ciphertext :

WECRL TEERDSOEEFEAOCAIVDED

Transposition (Rail-fence cipher) cipher :

W T O E V E
E L E S E F A I D N
C R E A C

Rail 1 : WTOEVE

Rail 2 : ELESEFAIDN

Rail 3 : CREAC

Decrypted message :

WE ARE DISCOVERED FLEE AT ONCE

Vulnerabilities of Transposition cipher

1. Preserved letter frequencies - Letter frequencies remains unchanged
2. Limited key Space
3. Known plaintext Attack
4. Weak security
5. No integrity Protection

Practical Enhancement :

Replace the classical cipher with AES-GCM

1. Encrypt the message using AES-GCM
2. Store the Encryption key securely
3. Used a unique nonce for Each Encryption under key.
4. verify the authetication before accepting the decrypted message.